

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2104

Bellville South 500 kV
April 26, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2104. The Interconnection Customer proposes a 235.4 MW solar plus storage facility interconnecting at the Bellville South 500 kV substation in the Illinois area. The request was received on June 03, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 507 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$645,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Napoleon Peaking Sta – Richland 230.0 kV L
Contingency: Troy Energy Llc – Richland 500.0 kV
Loading: 105.3% (Rating: 1,356 MVA)
DFAX: 12.7%
Pre-project loading: 103.1%
2. Facility: Gould – Station S Van Nuys 230.0 kV
Contingency: Troy Energy Llc – Richland 500.0 kV
Loading: 104.7% (Rating: 811 MVA)
DFAX: 6.4%
Pre-project loading: 102.8%
3. Facility: Wayne Trace – Darlington Rd 100.0 kV
Contingency: Richland – Dekalb 500.0 kV
Loading: 103.2% (Rating: 702 MVA)
DFAX: 5.8%
Pre-project loading: 101.3%
4. Facility: Ward Gulf – Laureles 100.0 kV
Contingency: Dekalb – Jones Chapel 500.0 kV line
Loading: 102.9% (Rating: 371 MVA)
DFAX: 5.2%
Pre-project loading: 99.6%
5. Facility: Landsman Crk – 24Th & Gardendale 230.0 kV
Contingency: Richland – Dekalb 500.0 kV
Loading: 102.0% (Rating: 1,003 MVA)
DFAX: 8.9%
Pre-project loading: 99.9%

6. Facility: Minnehaha – Sidney A Murray Jr Hydroelectric 230.0 kV
Contingency: Richland – Dekalb 500.0 kV
Loading: 101.5% (Rating: 1,750 MVA)
DFAX: 16.0%
Pre-project loading: 99.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Napoleon Peaking Sta – Richland 230.0 kV L	Reconductor line	\$7,654,000	2% (\$119,000)
Gould – Station S Van Nuys 230.0 kV	Reconductor line	\$5,165,000	2% (\$80,000)
Wayne Trace – Darlington Rd 100.0 kV	Reconductor line	\$642,000	2% (\$10,000)
Ward Gulf – Laureles 100.0 kV	Reconductor line	\$1,116,000	2% (\$17,000)
Landsman Crk – 24Th & Gardendale 230.0 kV	Reconductor line	\$15,630,000	2% (\$243,000)
Sidney A Murray Jr Hydroelectric – Minnehaha 230.0 kV line	Reconductor line	\$11,337,000	2% (\$176,000)
	Total allocated to project		\$645,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.